

COVER LETTER

August 24, 2026

To:

The Editor-in-Chief

Revista Portuguesa de Cardiologia (Portuguese Journal of Cardiology)

Elsevier Editorial System

Subject: Submission of an Original Investigation

Dear Editor-in-Chief,

Please consider our original research manuscript, “**Record-Level Evaluation of a Morlet CNN–Conformer for Five-Class ECG Beat Classification**,” for publication in *Revista Portuguesa de Cardiologia*.

Automatic electrocardiogram (ECG) beat classification and arrhythmia screening are cornerstone capabilities for modern cardiovascular telemonitoring, wearable cardiac devices, and hospital telemetry. However, common methodological pitfalls—notably random beat-level data splitting and opaque summary metrics—often mask severe model failures on critical minority arrhythmias (such as supraventricular ectopic beats and fusion beats).

In this work, we present a patient-independent, record-level evaluation of a 32-scale Morlet continuous wavelet transform coupled with a Conformer sequence encoder on 45 MIT–BIH records. While the Conformer architecture achieved superior aggregate performance (accuracy 0.60, macro-F1 0.26, weighted-F1 0.68) compared to CNN-attention, CNN-LSTM, and baseline CNN models, our failure-aware analysis shows that performance gains were concentrated almost exclusively in dominant classes (Normal and ventricular ectopic beats), with near-zero recognition of rare arrhythmias. We provide a thorough error matrix, threshold sensitivity, and calibration diagnostic to establish realistic boundaries for deep learning translation in clinical cardiology.

Author Declarations and Confirmations:

1. **Originality:** This manuscript is original, has not been published elsewhere, and is not concurrently submitted to any other journal.
2. **Authorship Approval:** All co-authors have reviewed, contributed to, and approved the final manuscript.
3. **Competing Interests:** The authors declare no financial or personal conflicts of interest.
4. **Funding:** This research received no specific grant from public, commercial, or non-profit funding bodies.
5. **Ethics:** Ethical approval and informed consent were not required as this computational study utilized the open-access, fully de-identified MIT–BIH dataset via PhysioNet.
6. **Reproducibility:** Complete source code, model checkpoints, and prediction arrays are publicly available in our versioned GitHub release.

Thank you very much for your consideration of our manuscript.

Sincerely yours,

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